MIL-S-1980D

19 January 1968
SUPERSEDING
MIL-S-1980C
14 August 1963

MILITARY SPECIFICATION

STAND, MIXING BOWL, MOBILE

This Specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one size of mobile mixing bowl stand for holding bakers' mixing bowls.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

*QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon
*TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-B-2316	-	Bakery Equipment, Including Unit Assemblies, Repair Parts and Tools, Preparation for Delivery of
*MIL-P-26915	-	Primer Coating, Zinc Dust Pigmented, For Steel Surfaces

FSC 7330

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U. S. Military Property

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply:

* NATIONAL SANITATION FOUNDATION (NSF)
Standard No. 2 - Food Service Equipment (1965)
Seal of Approval Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P. O. Box 1468, Ann Arbor, Michigan 48106).

* (Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 Preproduction sample.- Unless otherwise specified (see 6.2), before production is commenced, a sample mixing bowl stand shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The mixing bowl stand delivered under this specification shall be the manufacturer's commercial standard product, except for any changes necessary to comply with specification requirements.

* 3.3 Standard compliance.- The mixing bowl stand shall conform to the applicable requirements of NSF Standard No. 2.

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* 3.3.1 Certification.- Prior to approval of the preproduction sample, if one is submitted, or prior to the approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the mixing bowl stand he proposes to supply under this specification meets the applicable requirements of NSF Standard No. 2. Acceptable evidence of meeting the requirements of this standard shall be the NSF Seal on the stand, a listing in the NSF "Seal of Approval Listing of Food Service Equipment", or a certified test report from a recongized independent testing laboratory acceptable to the medical service of the procuring activity indicating that the mixing bowl stand offered conforms to applicable requirements of NSF Standard No. 2.

3.4 Materials.- Materials not definitely specified shall be of the quality normally used in the manufacturer's standard product. Commercial mill tolerances are applicable to all material thicknesses.

* 3.4.1 Carbon steel.- Carbon steel shall conform to the chemical composition of any of the steel numbers from 1009 to 1020 inclusive, as specified in QQ-S-698. Steel shall be hot rolled (HRCQ) or cold rolled, commercial quality (CRCQ).

* 3.4.2 Zinc dust primer.- Zinc dust primer shall conform to MIL-P-26915.

3.5 Design and construction.- The mixing bowl stand and reducing ring shall be fabricated of zinc-coated carbon steel or carbon steel specified in 3.4.1 which is galvanzied after fabrication. The mixing bowl stand shall support a load of 200 pounds in a mixing bowl without tipping, displacement of bowl, or failure or deformation of any component when rolled over a 3/8 inch high obstacle (see 4.6.1).

3.5.1 Body.- The body shall be fabricated from carbon steel of the composition specified in 3.4.1 and shall incorporate an upper circular band and a lower circular band or rectangular base. Bands shall be 0.1793 inch thick by 2 inches wide. Four uprights, equally spaced within plus or minus 1/8 inch around the upper and lower band or base, shall be provided. The uprights shall be 0.1793 inch thick by 2 inches wide flat carbon steel specified in 3.4.1 when a circular lower band is provided, and angle iron when a rectangular base is provided. The uprights for the rectangular base shall be located at the corners, and all uprights shall be welded or riveted to the lower band or base. The uprights shall support the upper band and be welded or riveted thereto. The upper band shall have an inside diameter of 23-1/4 inches, plus or minus 1/8 inch. The stand may be provided with a carbon steel outer skin enclosing the bands and uprights. The overall height from the floor, including casters, shall be not less than 26 inches nor more than 28-5/8 inches.

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* 3.5.2 Casters.- The stand shall be provided with four, removable, swivel casters. One caster shall be located below each upright and shall not protrude outside of the base or lower band. Casters shall have either single or double ball bearing swivels and oilite, nylon, or graphite impregnated wheel bearings. Wheels shall be molded hard rubber (nonmarking) vulcanized permanently to a steel core or hub and shall be not less than 3 inches in diameter. The caster top plate shall be rectangular and shall be bolted to the body. Casters shall be capable of supporting the load specified in 4.6.1 without deformation or floor marking caused by rubber and shall roll and swivel freely.

3.5.3 Brakes.- The stand shall be equipped with either a foot operated brake or two casters shall be equipped with brakes. The foot operated brake shall be a hinged foot plate capable of being swung in-and-up when not in use and down when braking. When in the braking position, the foot plate shall be positioned so that pressure by the foot shall depress the plate against the floor and prevent the stand from moving. When tested as specified in 4.6.2 the stand shall not move laterally.

3.5.4 Reducing ring.- A reducing ring for holding smaller bowls shall be provided with each stand. The ring shall have an inside diameter of 20-1/2 inches, plus or minus 1/8 inch. The outside diameter of the reducing ring shall fit inside the upper band and shall be supported in place by not less than three equally spaced (plus or minus 1/8 inch) lugs or clips, welded or riveted to the reducing ring.

* 3.6 Finish.- When not fabricated of zinc-coated steel, the stand and reducing ring (see 3.6.1) shall be galvanized after complete fabrication. Galvanizing shall be accomplished by electroplating, hot-dipping, or by use of zinc-dust primer specified in 3.4.2. When fabricated of zinc-coated steel, the surfaces from which zinc has been removed during fabrication shall be recoated. Electroplated coating shall be minimum 0.00015 inch thick and hot-dipped coating shall be minimum 0.003 inch thick. When zinc-dust primer is used, the stand and ring shall be cleaned and prepared for painting in accordance with any method specified in TT-C-490 followed by coating in accordance with MIL-P-26915. Primer finish shall be uniform in color and appearance and free from runs, sags, or wrinkles. Galvanizing shall be clean and free from lumps, blisters, and bare spots.

* 3.6.1 Hardware.- Hardware and exposed metal parts of casters shall be zinc-coated, cadmium plated, or be corrosion-resisting steel.

3.7 Marking for identification.- The mixing bowl stand shall be marked in accordance with MIL-STD-130.

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* 3.8 Workmanship.- Rough and sharp edges, burrs, and slivers shall be removed. Welds shall be sound, smooth, and free from pits, burn through, or fissures. Rivets shall be tightly driven and fully peened. Bolts and nuts shall be tightly drawn and lockwashers shall be provided.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection.- When a preproduction sample is required, it shall be examined for the defects in table I, dimensions specified herein, and shall be tested as specified in 4.6.1 and 4.6.2.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Process examination.- Examination shall be made of galvanizing thickness to determine conformance to 3.6. Whenever nonconformance is noted, correction shall be made to process and affected items.

4.3.3 End item inspection.- The inspection lot shall consist of all stands offered for inspection at one time. The sample unit shall be one completely fabricated mixing stand.

4.3.3.1 Visual examination.- Examination of the end item shall be in accordance with table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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* TABLE I.- Classification of defects.

Examine	Defect	Classification Major	Minor
Finish	Stand and reducing ring not galvanized or not fabricated from zinc-coated steel	X	
	Surface not clean or contains lumps, blisters, runs, bare spots, sags, or wrinkles		X
Construction and workmanship	Burrs, slivers, and sharp edges not removed	X	
	Any component fractured or malformed	X	
	Any part missing or not specified type	X	
	Any part loose		X
	Any part not properly aligned	X	
Casters	Not removable	X	
	Not equipped with brakes (when applicable)	X	
	Less than two casters equipped with brakes (when applicable)	X	
Foot lever (when applicable)	Foot lever does not rest on floor (in braking position)	X	
	Hinge binds		X

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* TABLE I.- Classification of defects. (cont'd)

Examine	Defect	Classification Major	Minor
Reducing ring	Binds in band		X
	Less than four lugs or clips	X	
Welding	Missing, incomplete fusion, burn-holes, cracks, undercut, or not otherwise fused	X	
	Slag inclusion, not smooth and uniform, scale or flux deposits not removed		X
Rivets (when applicable)	Rivet loose or not fully peened		X
	Rivet split	X	
Threaded fastener	Stripped		X
Identification marking	Missing, incomplete, not legible		X

* 4.3.3.2 Dimensional examination.- Examination shall be made of the end item to determine conformance to dimensional requirements of this specification. Any nonconformance to specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

* 4.3.3.3 Testing of the end item.- When a preproduction sample is not required, the initial production unit shall be tested as specified in 4.6.1 and 4.6.2. In addition the end item shall be tested as specified in 4.6.1 and 4.6.2 with an inspection level of S-4 and an AQL of 4.0 defects, expressed in terms of defects per hundred units.

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* 4.4 Examination of preparation for delivery.- Examination shall be made of preservation, packaging, packing, and marking in accordance with MIL-B-2316.

4.5 Standards compliance.- Proof of compliance with NSF requirements of 3.3 shall be available to the Government representative.

4.6 Tests.-

4.6.1 Structural test.- The completely assembled stand shall be loaded with a mixing bowl containing 200 pounds, plus or minus 5 pounds, then pushed ten times over a 3-inch wide floor obstacle, three-eighths of an inch high having 90-degree faces, with longitudinal axis of the obstacle set at an angle of 30 degrees to the direction of travel of the stand. During the test the stand shall not tip over, and after the test the stand, casters, and floor surface shall be examined to determine there is no permanent deformation or floor marking as specified in 3.5 and 3.5.2.

4.6.2 Brake test.- With the load still applied, the foot operated brake shall be depressed or two brake equipped casters shall be engaged and a 50-pound horizontal force successively applied at each upright to determine conformance to 3.5.3.

* 5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C.- Preservation, packaging, packing, and marking for each level of protection shall be made in accordance with the applicable requirements of MIL-B-2316 (see 6.2).

6. NOTES

6.1 Intended use.- The mixing bowl stand covered by this specification is intended to be used for moving food ingredients in food mixing bowls to and from mixing machines in permanent bakeries and kitchens.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of the specification.
(b) When preproduction sample is not required (see 3.1).
(c) Selection of the applicable level of preservation and packaging and packing (see 5.1).

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6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

Army - GL

Project No. 7330-0245

Review activities:

Army - MD
Navy - SA
Air Force - 84, 85

User activity:

Navy - MC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Stand, Mixing Bowl, Mobile - MIL-S-1980D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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